

THERIF: Themes for Readability from Iterative Feedback

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ABSTRACT

Digital reading applications give readers the ability to customize fonts, sizes, and spacings, all of which have been shown to improve the reading experience for readers from different demographics. However, tweaking these text features can be challenging, especially given their interactions on the final look and feel of the text. Our solution is to offer readers preset combinations of font, character, word and line spacing, which we bundle together into reading themes. To arrive at a recommended set of reading themes, we combine crowdsourced text adjustments, ML-driven clustering of the resulting text formats, and design sessions. After four iterations of these steps, we converge on a set of three COR (Compact, Open, and Relaxed) themes that are designed for different readers.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing; Interactive systems and tools**; • **Computing methodologies** → *Cluster analysis*.

KEYWORDS

human-in-the-loop, reading, presets

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1 INTRODUCTION

We consume most of our information in digital form: news on mobile devices, work emails on our computers, and leisurely reading on our e-readers. Customization of text formats including the presence of serifs and stroke widths have been shown to differently affect readers with dyslexia, readers of old age, children learning to read, etc. [5, 15, 18, 29, 31, 35, 43, 67, 78]. Large reading gains are also possible by individuating font and other text characteristics to

each reader [6, 8, 10, 16–23, 61, 63, 64, 68, 74, 80, 81]. Increasingly, applications we use afford users control of text settings. However, aside from increasing text size to be more legible, readers may not know how to adjust text settings to their needs. Moreover, many of the settings are interrelated, where adjustments to character spacing, for example, may require further adjustments to word or line spacing to feel comfortable. This is a difficult text formatting problem to leave in the hands of casual readers.

To close this gap and offer readers comfortable reading formats as starting points for further customization, we bundle fonts and spacings together into **reading themes**. Recognizing that we want the reading themes to both fit diverse readers' preferences and be well-suited for future use in reading applications, we follow established approaches in crowdsourcing design and inclusive design guidelines to generate **themes** for readability with iterative feedback (**THERIF**). Using this approach, we continually iterate through crowdsourced text setting refinements, automatic clustering, and design sessions (Figure 1). With a focus on English reading, we show that multiple iterations of crowdsourced theme refinement can yield reading themes that are well-suited for different readers.¹

2 RELATED WORK

Typographic characteristics affect digital reading. A wide range of typographic characteristics have been shown to affect individual comfort and performance when reading digitally. Font choice can significantly affect readability [6, 8, 10, 16–20, 61, 63, 68, 81], which may be attributed to characteristics such as font weight [55], stroke contrast [9, 11, 12], and character width [47, 57]. The effects of spacings on digital reading similarly vary by reader: larger character spacing may benefit readers with dyslexia [45, 64, 88], with low vision [12], and those reading unfamiliar content [78], but it may also hinder proficient readers [42]. Past work recommended increasing word spacing proportionally with character spacing to avoid compromising reading performance [32, 69]. To our knowledge, no consistent recommendation for line spacing exists, but previous work and guidelines often recommended maintaining at least a single line spacing [30, 41, 68].

Personalizing text settings is challenging. Many recommendations exist for tailoring text settings to individuals, but adopting them can be challenging for casual readers. Reader characteristics

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¹In the full paper available at <https://arxiv.org/abs/2303.04221>, we demonstrate the study design and results in greater detail.

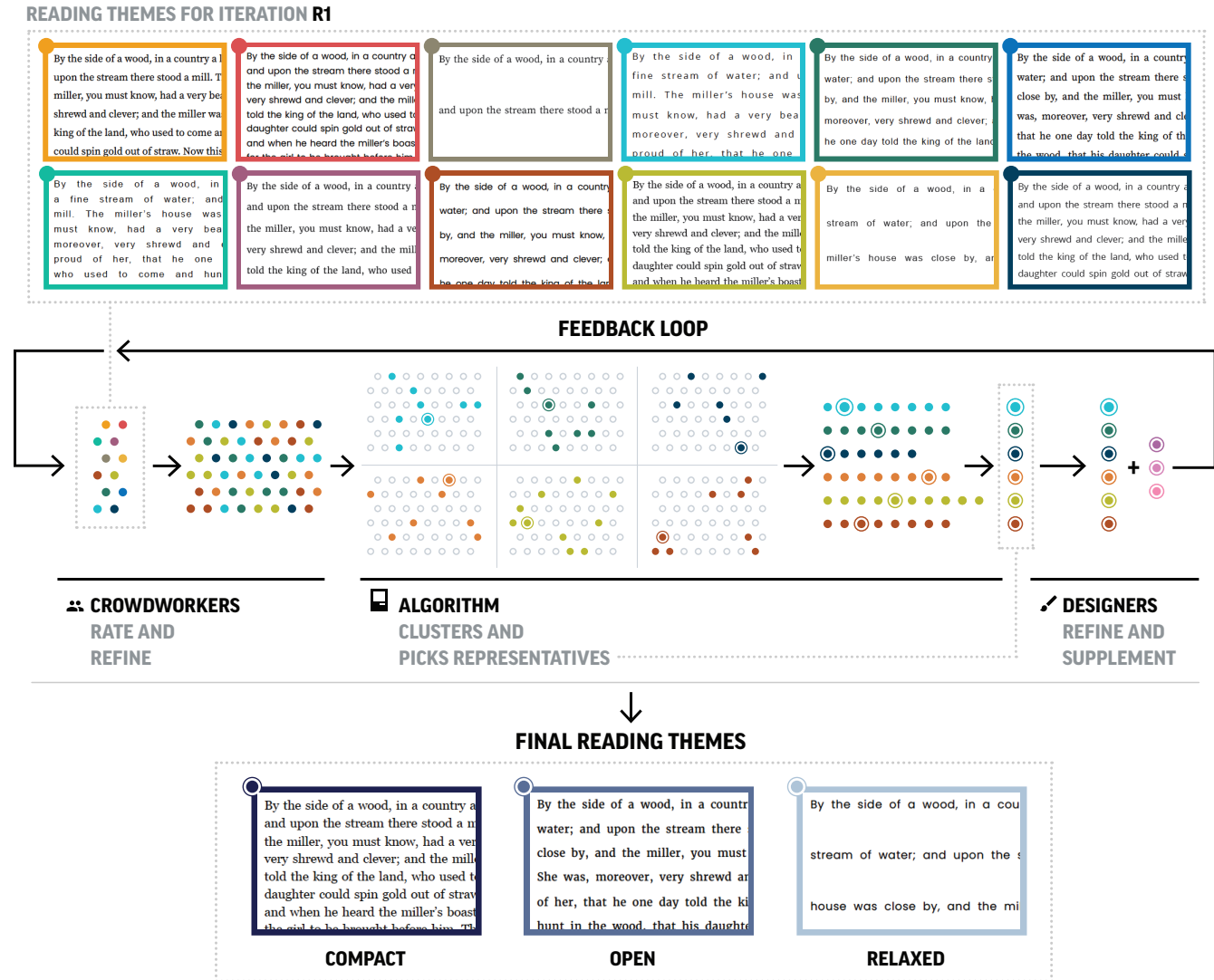


Figure 1: Our pipeline for generating themes for readability from iterative feedback (THERIF). Each iteration is initialized with text presets bundled into reading themes that crowdworkers rate and refine. The resulting customized text formats are then automatically clustered using an ML-based algorithm, and cluster representatives are selected as reading themes for the next iteration. Designers refine and supplement the automatically generated themes with additional themes before the iterative loop continues with a new set of crowdworkers. Each colored dot represents a customized text format, with variations to font, as well as character, word, and line spacing. Crops from example text formats produced after a single iteration of THERIF are displayed at the top. On the bottom are the three COR (Compact, Open, Relaxed) reading themes obtained after four iterations of THERIF.

such as dyslexia and language fluency vary on a continuous spectrum [25, 75], but recommendations for text settings are largely made based on demographic categories (e.g., readers with dyslexia, children, etc.), without consideration for overlap in reader needs and reader characteristics that vary on a continuum. Additionally, recommendations often do not account for relevant contexts, such as time of day, type of reading, or situationally-induced impairments and disabilities (SIIDs) [27, 40, 85].

Presets can simplify the space of options. Guiding readers but leaving them with agency over the final reading formats may offer greater flexibility to their local context [14, 25, 27, 40, 75, 85]. However, the reading and text setting interfaces available today are not well-suited for this purpose, as continuous adjustments to spacing and long drop-down menus of fonts are challenging for casual readers to navigate [35, 56]. These problems point to the need to guide readers towards *presets*, and give them good defaults, or starting points for further customization [1, 34, 51, 58].

By offering multiple presets, designers can tailor the same interface to groups of users with different preferences. Leveraging these learnings, we developed presets of text settings that we refer to as **reading themes**.

Crowdsourced design frameworks capture user preferences. The participatory design framework involves designer-user collaborations to create designs that effectively capture user preferences [77]. This framework is well-suited for the design of reading themes where diverse user preferences need to be represented and alignment with typographical guidelines are similarly important. Previous works explored large-scale user participation in individual design steps [26, 72, 84], and in the entire design process [59, 86]. Crowdworkers have shown that they can collectively create designs with high originality and quality [24, 54, 86], and such input is especially relevant given the diversity of reading needs.

3 SELECTING A REDUCED SET OF TEXT SETTINGS

Reading applications offer control over different text settings such as font choice and size, character, word and line spacing. However, the space of possible text adjustment settings is large, and systematically iterating over combinations of these text features would be intractable. We leverage the volume and diversity of crowdworkers to sample this space based on their preference. For this purpose, we built a prototype that offers readers fine-grained control over their text settings, and put this **text settings prototype** in front of crowdworkers to discover which settings are most commonly adjusted and used together to arrive at their preferred reading formats. In this section, we describe our prototype and how its design evolved with learnings from the pilot study. The final prototype design was used for generating our reading themes (§4). Data from the pilot study were also used to initialize the reading themes in the main study.

Initial text settings. The first version of our text settings prototype offered participants an exhaustive list of 11 text adjustments based on prior literature [35, 48, 49], including fonts, sizes, spacings, text alignments, and color themes, among others (Appendix Figure 5). We offered eight fonts: Montserrat, Open Sans, Arial, Roboto, Merriweather, Georgia, Source Serif Pro, and Times, based on previous work reporting they are diverse, prevalent, and readable, with characteristics generalizable to other fonts [21]. Participants could adjust the settings in the **text settings panel** and preview changes to the text in a **reading panel** preloaded with long-form English passages. All text settings included wide ranges of possible values and supported adjustments in both step increments and with a continuous slider. In later parts of our study, a **theme review panel** also allowed participants to review and rate text setting presets (Figure 2a).

Pilot study. We conducted a pilot study with the prototype composed of four iterations conducted on the UserTesting platform, using the think-aloud protocol to collect participants' feedback. The study involved a total of 271 crowdworkers and design sessions with four designers: D1: Man with 27 years of design experience; D2: Man with 20 years of design and typography experience; D3: Woman with 10 years of design experience; D4: Woman with 8

years of design experience.² Iterations included 100, 51, 60, and 60 crowdworkers, respectively. Studies lasted an average of 40 minutes, with each participant compensated 30 USD hourly. In each iteration, crowdworkers used the text settings panel to design their preferred reading formats, and then the designers (in sessions of 30–45 minutes) reviewed the results from an automatic clustering algorithm that grouped similar reading formats together, and helped select representative reading themes for the next iteration (§4.2).

Learnings. Two authors used descriptive codes [71] to summarize participants' design processes and think-aloud feedback, and identify frequently used text settings and adjustment patterns. The authors then met with all four designers to review the descriptive codes, relate them to participants' designs, and develop high-level learnings. We incorporated these learnings to refine both the prototype and main study design:

- We only kept font choice and spacing settings (Appendix Table 1). Other settings were either too context-dependent to be bundled into themes (e.g., background color or contrast) or considered irrelevant for readability by designers or previous work (e.g., paragraph indentation or spacing) [3, 7, 28, 39, 48, 62].
- A participant's viewing distance and screen resolution may confound the optimal font size [43]. Based on the designers' suggestion, we fixed the font size in the prototype such that all fonts have the same x-heights. The intention is for font size to be set independently from the reading themes.
- To facilitate exploration of a greater range of text settings, we used the crowdsourced text settings from the pilot study to initialize the prototype in the main study.³
- We randomized the order of fonts in the dropdown list to mitigate positional bias and encourage exploration.

4 INTRODUCING THERIF

In this section, we introduce our pipeline for generating **themes** for readability with iterative feedback, which we refer to as THERIF. Each iteration of THERIF included three stages: (1) Crowdworkers customized text settings to create diverse reading formats (§4.1); (2) ML algorithms automatically clustered the resulting formats and designated cluster representatives as *reading themes* for the next iteration (§4.2); (3) Designers validated the automatically selected themes and created additional reading themes (§4.3). Stage 1 was motivated by user-centered design and participatory design practices, and prior work demonstrating the successful use of crowdsourcing for design studies [24, 26, 54, 59, 72, 84, 86]. Stage 2 leveraged machine learning to simplify collaboration between the crowd and the expert [36]. Stage 3 was intended to ensure good design practices were followed in the design of readable text formats.

²All designers were recruited from the same large U.S. corporation. Only full-time work experience was reported. Designers were similar to the population they design for [53, 82]. Based on their responses to the dyslexia questionnaire [25], two designers had above-average chances of having dyslexia.

³In the pilot study, we started all participants with the default setting of Arial font at 16px, 1.2 line spacing, and default values of character and word spacing (0em), and we asked them to explore the text settings to arrive at a preferred format.

THERIF iterations. We refer to the THERIF iterations by R0, R1, R2, and R3. Each iteration starts with a group of crowdworkers adjusting text settings from the provided defaults to their liking. The very first iteration (R0) is initialized with themes from the pilot study. All subsequent iterations (R1-R3) are initialized with themes from the previous iteration, after automatic clustering and design sessions. While we ran four iterations to produce the final set of themes in this paper, this pipeline is extensible to future iterations. For every iteration, we recruit a new set of crowdworkers that have not participated in this study before, to ensure that the reading themes evolve to be representative of different readers' preferences rather than fine-tuned to the preferences of a few.

Crowdworkers. We used the Prolific platform to recruit crowdworkers with English as their first language. We recruited 200 participants for the first iteration (R0) and 100 for each subsequent iteration (R1-R3), for a total of 500 participants.⁴ Dyslexia and age are important factors requiring interface adaptations when reading [15, 21, 22, 44, 48, 63–66, 81]. In each iteration, we recruited roughly 50% participants with dyslexia and 50% without.⁵ See Appendix B.1 for how we identify readers with dyslexia and Appendix B for more details on participant demographics.

4.1 Stage 1: Crowdworkers rate and refine

Every THERIF iteration starts with a new population of crowdworkers using the reading settings prototype (Figure 2) to refine the text settings based on their preference. The prototype is initialized with the set of reading themes from the previous iteration of THERIF.⁶ Crowdworkers are guided through the following steps:

Review themes. Crowdworkers in THERIF iterations R1-R3 were instructed to first review and rate each theme (good, unsure, or bad) from the previous iteration before selecting their favorite theme (Figure 2a). We always included a randomly selected **validation theme**, intended to represent a poor reading format.⁷

Refine text settings. Starting with their preferred theme (iterations R1-R3) or a randomly initialized theme (R0), crowdworkers were then instructed to toggle on the text settings panel (Figure 2b) and try out each setting to observe how it affected the text using a think-aloud protocol. This familiarized them with the text settings.

Repeat for final selections. After completing the steps above and getting acquainted with the text settings, we reset the interface and asked crowdworkers to repeat both steps and customize their reading theme to their liking. Repeating the steps allowed crowdworkers to select a theme knowing further refinements were possible.

⁴The larger number of participants in R0 was to allow the presets to deviate from the reading themes initialized from the pilot study, thus increasing the diversity of the formats in future iterations.

⁵Readers with symptoms of dyslexia account for about 15-20% of the world population [4]. We maintained equal representation to ensure that their reading preferences were also adequately considered.

⁶The very first iteration (R0) is initialized with a single theme for each participant, randomly selected from 6 representative combinations of (*character, word, line spacings*) from the pilot study (§3), randomly paired with one of our 8 study fonts to offer diverse starting points.

⁷Eleven validation themes were selected from the pilot study by designers asked to spot poor reading formats created by crowdworkers.

4.2 Stage 2: Algorithm clusters and picks representatives

To converge on a handful of reading themes that represent different reader preferences, we used a convolutional neural network and k-Means algorithm (Appendix C) to cluster all the reading formats generated by crowdworkers in Stage 1 of THERIF, and **select cluster representatives to serve as reading themes for the next iteration**. Since hundreds of different readers can arrive at similar formats, reducing this set of formats to representative examples makes evaluation of different format choices more tractable [36]. During the pilot study, we asked designers to help select cluster representatives, but this was a time-consuming process that did not achieve consensus across designers. Because we used a reduced set of text settings in the main study, automatically selected cluster centroids by the k-Means algorithm were good representatives; when shown to designers for validation, they agreed they would have made similar selections.

4.3 Stage 3: Designers refine and supplement

Document formatting, especially getting the combination of fonts and spacings to look and feel right, is an involved design task. During the pilot design sessions, D2 commented that designing reading formats that respond to different people “compound the variables designers have to consider”, including but not limited to “glyph characteristics and a variety of spacing features”. To incorporate design considerations into the theme feedback loop, we asked designers to evaluate the reading themes selected after automatic clustering by allowing them to view each theme and refine it further using the text settings panel. Designers could then save their refinements as additional themes. We used the designers from the pilot study. To keep the task tractable, one designer reviewed the themes after each iteration: D1 added 3 themes after R0, D2 added 6 themes after R1, and D3 added 3 themes after R2. In the last iteration (R3), we chose not to involve designers in creating additional themes or modifying the existing ones, to directly evaluate the crowdsourced themes generated by THERIF. The themes generated from the automatic clustering supplemented by the designers' themes formed the full set of reading themes presented to crowdworkers on each THERIF iteration (12 themes in R1, 12 in R2, and 6 in R3). By involving both designers and new sets of crowdsourced participants in evaluating themes generated after each iteration, we continued improving the themes over time.

5 FINAL THEMES

Each iteration of THERIF resulted in a set of reading themes, obtained by clustering crowdsourced text adjustments. After four iterations of THERIF, three clusters remained, and the cluster representatives are our final three reading themes, without any further designer adjustments. We call them the **COR themes: Compact, Open, Relaxed**. We visualize the themes in Figure 3 and include CSS to reproduce them in Appendix D. Each of the themes represents a distinct reading format. Themes with larger line spacing naturally ended up with larger character and word spacing (across the THERIF iterations), corresponding to common typographical considerations and past research [38, 69].

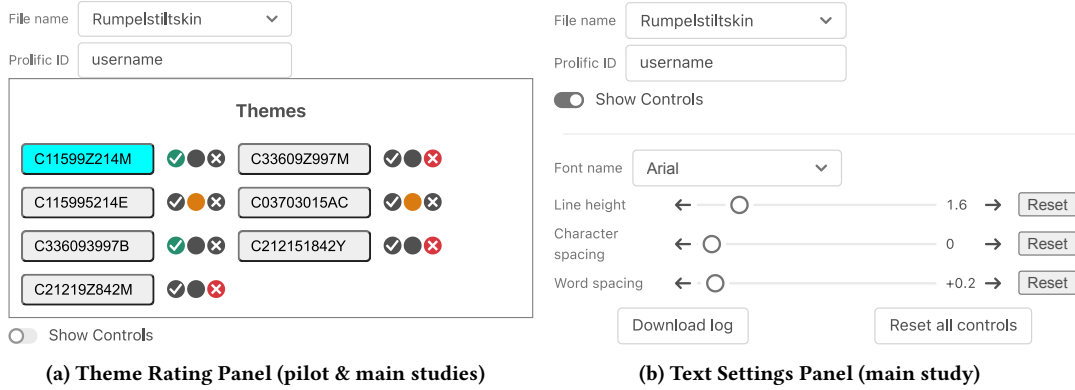


Figure 2: The left panel used for the main study includes views for theme rating and refinement. Participants first rate each theme (good, unsure, bad) and select their preferred reading theme whenever applicable (a). Once finished, they were instructed to toggle on “Show Control” to expose the text settings and make further refinements (b). In the main study, the left control panel hides when the mouse moves away, allowing participants better focus on the reading format. The theme names are anonymized alphanumerical IDs.

The fonts in our reading themes agree with previous work showing they correspond to either preferences or good reading performance. Cai et al. used the same eight fonts as in this study and reported that Merriweather (Open theme) was the most preferred font by participants [21], Georgia (Compact theme) resulted in the largest improvement in reading performance, and Poppins (Relaxed theme) led to speed improvements, especially for language learners [21, 42].

We include demographics of crowdworkers whose text formats were clustered together to produce the COR reading themes (Figure 3). We observed that more participants ended up in the cluster with larger spacing, although 14% of participants preferred the most compact reading format. Both Open and Relaxed themes conform to the Web Content Accessibility Guidelines for word and line spacing [41]. Though the settings of the Compact theme are slightly below the recommendation, they are nonetheless above modern browsers’ default settings [52]. Most of the participants preferring the compact setting were 26-45, representing the portion of the population more likely to be working professionals. In our pilot study, we found that this cohort of the population was more likely to read for work and have to navigate text quickly, so a more compact format could suit these needs. Participants older than 55 seldom chose a more compact text setting. Formats preferred by participants with dyslexia were more likely part of the reading theme with the largest spacing, supported also by previous literature [64]. Noteworthy is that this format was also preferred by a number of participants without dyslexia.

6 THEMES CONVERGE TO MATCH READERS’ PREFERENCES

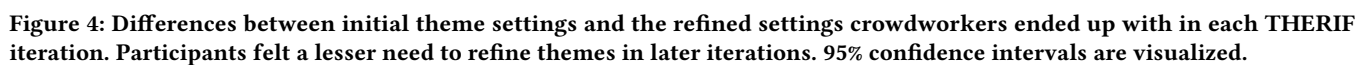
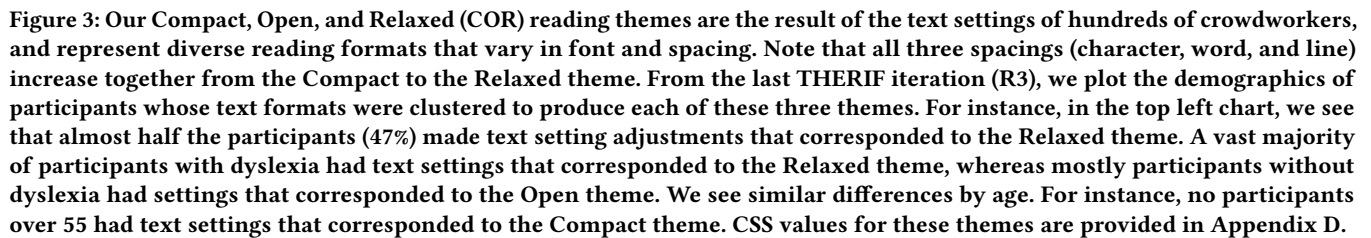
During the initial refinement iteration (R0), we observed that all participants increased character, word, and line spacing from the default theme, resulting in a set of diverse formats (Figure 4). Participants with dyslexia increased word spacing during R1 and R3, and participants without dyslexia increased character spacing in

R2 and word spacing in R3, respectively (Appendix Table 3). As the THERIF iterations progressed, participants made fewer refinements to the provided themes (Figure 4), indicating that the defaults were “good enough”, i.e., meeting their preferences.

Apart from examining the text settings themselves, we also considered the time participants spent making adjustments to the default themes. Comparing iterations, we found that participants in R1 spent significantly less time on refinement ($M = 34$ sec) than those in R0 ($M = 50$ sec, $t(272.39) = -2.42$, $p = 0.02$, Cohen’s $d = -0.26$). Refinement time may be related to the number of themes presented. The fewer theme defaults available, the more likely participants will spend more time making refinements, which explains why participants in R0 with one default setting spent the most time customizing their reading format. However, despite the decreasing number of themes from 12 in R1 to 6 in R3, participants did not spend more time on refinements. No significant difference existed when comparing the refinement time between R1 and R2 ($M = 30$ sec) or R2 and R3 ($M = 39$ sec; Appendix Figure 7), indicating that the smaller number of themes were nevertheless meeting participant preferences.

7 CONCLUSION

Digital reading applications available today occasionally offer readers custom control over certain text settings like font, size, or spacing. Despite the benefits of personalization on reading performance [21, 23, 46, 68, 80, 87], casual readers may still find adjusting these text settings cumbersome. To this end, we propose providing readers with reading themes: preset combinations of fonts and spacings that can be starting points for further customization. Indeed, our analyses show that as we refined the themes from one THERIF iteration to the next, crowdworkers made fewer and fewer adjustments to the themes they were provided with, demonstrating that our themes had converged to better represent readers’ preferences. By recruiting new crowdworkers in each THERIF iteration, equally sampling readers with and without dyslexia, and covering a wide



range of ages, our final themes naturally incorporate diverse reader preferences, without directly mapping reading themes/settings to demographic categories. We believe that reader needs, just like reader characteristics, vary on a continuum. This work brings us a step closer to allowing every reader, struggling or proficient, young or old, to read comfortably. Where most reading today occurs on digital surfaces, text that caters to individual reader needs should be the rule, not the exception. This is where customization and inclusivity go hand-in-hand.

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A TEXT SETTINGS IN PILOT AND MAIN STUDIES

Table 1 compares text settings included in the pilot and the main study. See Figure 5 for the pilot study interface.

Text Settings	Pilot	Main
Character Spacing	+	+
Word Spacing	+	+
Line Spacing	+	+
Font Name	+	+
Font Size	+	-
Paragraph Indent	+	-
Paragraph Spacing	+	-
Column Width	+	-
Text Alignment	+	-
Color and Contrast	+	-
Dark Mode	+	-

Table 1: The main study includes a subset of the text settings used in the pilot study. Settings marked “+” could be adjusted by the participants in the main study, and those marked “-” were fixed after the pilot study. The pilot study helped us identify which settings lead to systematically different preferences across participants. Based on participant and designer feedback, we removed settings unrelated to readability (e.g., paragraph spacing) or that vary considerably across reading contexts (e.g., dark mode). During the main study, all fonts were shown at the same x-height as 17px Times, paragraphs had no indent, paragraphs each had 1em spacing before and after, and the column width was 6in. All texts were left-aligned, in black, over a white background.

B DEMOGRAPHICS

We recruited participants without dyslexia equally from different age brackets: 18-25, 26-35, 36-45, 46-55, and 56-87.⁸ However, when recruiting participants with dyslexia, we were unable to recruit equally from each age group due to the limited number of participants with dyslexia on the crowdsourcing platform. Nonetheless, compared to previous readability studies [44, 64, 66], our study has significantly more participants with dyslexia, a more balanced representation across age groups, and overall, a larger number of participants. These participants are not intended to be a representative sample of the population in age and dyslexia, but instead to explicitly include diverse readers.

For data quality purposes, we removed participants who (1) failed the attention check question (“Do you bike across the pacific to get to work each day?”), (2) did not complete the full study, (3) did not correctly fill in their username, or (4) finished the study exceptionally fast (three standard deviations faster than the mean of their age group). 485 participants remained after data removal, 237 (49%) with dyslexia and 248 (51%) without. Among the participants, 287

⁸The study lasted eight months due to the need to recruit sufficient number of readers with dyslexia in each design iteration. Participants’ age information reflects their age at the time of data export rather than the time of the study.

File name: Rumpelstiltskin

Font name: Arial

Font size: 16

Line height: 1.2

Character spacing: 0

Word spacing: +0.2

Paragraph indent: 0

Paragraph spacing: 0

Text alignment: Left

Column width: 6

☐ Dark mode

Color theme: Abc Abc Abc Abc Abc Abc
Abc Abc Abc Abc Abc Abc

Contrast: 90

☐ Show reading ruler

Download log Reset all controls

(a) Text Settings Panel (pilot study)

RUMPELSTILTSKIN

By the side of a wood, in a country a long way off, ran a fine stream of water; and upon the stream there stood a mill. The miller's house was close by, and the miller, you must know, had a very beautiful daughter. She was, moreover, very shrewd and clever; and the miller was so proud of her, that he one day told the king of the land, who used to come and hunt in the wood, that his daughter could spin gold out of straw. Now this king was very fond of money; and when he heard the miller's boast his greediness was raised, and he sent for the girl to be brought before him. Then he led her to a chamber in his palace where there was a great heap of straw, and gave her a spinning-wheel, and said, 'All this must be spun into gold before morning, as you love your life.' It was in vain that the poor maiden said that it was only a silly boast of her father, for that she could do no such thing as spin straw into gold: the chamber door was locked, and she was left alone.

She sat down in one corner of the room, and began to bewail her hard fate; when on a sudden the door opened, and a droll-looking little man hobbled in, and said, 'Good morrow to you, my good lass; what are you weeping for?' 'Alas!' said she, 'I must spin this straw into gold, and I know not how.' 'What will you give me,' said the hobgoblin, 'to do it for you?' 'My necklace,' replied the maiden. He took her at her word, and sat himself down to the wheel, and whistled and sang:

'Round about, round about,

Lo and behold!

Reel away, reel away, Straw into gold!

And round about the wheel went merrily; the work was quickly done, and the straw was all spun into gold.

When the king came and saw this, he was greatly astonished and pleased; but his heart grew still more greedy of gain, and he shut up the poor miller's daughter again with a fresh task. Then she knew not what to do, and sat down once more to weep; but the dwarf soon opened the door, and said, 'What will you give me to do your task?' 'The ring on my finger,' said she. So her little friend took the ring, and began to work at the wheel again, and whistled and sang:

(b) Reading Panel (pilot & main studies)

Figure 5: The left and right panels of the study interface used during the pilot study. The left panel (a) offers control over a comprehensive set of text settings, and the right panel (b) shows a preview of the reading format based on the current settings. Participants can view the effect of their text settings on four different passages.

(59%) are women, 196 (40%) are men, and 2 (< 1%) did not answer. Studies lasted around 30 minutes on average, and participants were compensated 13.5 USD hourly.

B.1 Identifying readers with dyslexia

Identifying participants with dyslexia is challenging because access to comprehensive assessments for dyslexia can be expensive [13, 76], and many with language difficulties go undiagnosed [2, 33, 50, 60]. To mitigate these challenges, we identified participants with a higher than average chance of having dyslexia using a questionnaire developed by Cooper and Miles [25]. This questionnaire has shown effectiveness at identifying readers with dyslexia and was adopted by past readability studies [37, 75, 83, 88]. We conducted a screening study with 1,608 Prolific participants who reported having difficulty reading. Among them, we identified 397 (25%) participants who may have dyslexia, and 237 took part in our study. *Disclaimer:* In addition to those diagnosed with dyslexia, we refer to participants who scored highly on the dyslexia questionnaire as "participants with dyslexia" in this paper, although they may not have been formally diagnosed nor have self-identified as such.

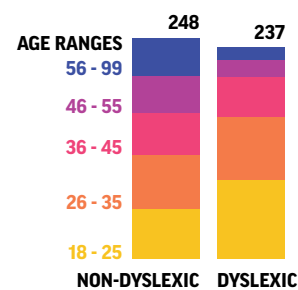


Figure 6: A total of 485 crowdworkers participated in the main study (R0-R3). We recruited equal numbers of participants with and without dyslexia and attempted to balance participants across ages by recruiting separately from each of the age buckets visualized. This was easier to do for participants without dyslexia, but participants with dyslexia were scarcer on the crowdsourcing platform.

C CLUSTER CROWDSOURCED READING FORMATS WITH CNN AND K-MEANS

We tried a few different approaches to cluster similar reading formats together. In one approach, we used the values of the text settings (fonts, spacings) as the features for clustering. However, it is not clear how to weight these different features, as they have different effects on the final text appearance. Instead, we found that more interpretable clusters formed when we used screenshots of the reading formats.

We trained a convolutional neural network (CNN) on crops of reading format to group similar reading formats together in a self-supervised way. We reproduced screenshots of participants' reading formats from their refinements log files (similar to the examples in Figure 1; see supplementary material for real samples). We then used random crops of the screenshots to train a CNN to predict the source (participant ID) of each crop, i.e., a self-supervised training approach. This ensured that the CNN model learned to associate crops from the same reading formats, which would allow it to later group similar formats together. Diversity in participant preferences made self-supervised training viable. The model was trained on data from R0, where 191 participants created 174 (91%) distinct reading formats. We experimented with the crops and selected a size (128px per side, equivalent to 3.36 deg of visual angle) that captured multiple lines of text at varied spacings. The final model achieved an accuracy of 79% on the test set. We then used the feature vectors from the penultimate layer of the trained CNN for clustering.

We used the trained CNN to produce feature vectors for the reading formats, which we then cluster using the k-Means algorithm [79]. We ran the k-Means algorithm on feature vectors of each crop to group together similar formats designed by different participants [79]. We used 1000 random crops from each reading format to thoroughly represent the format and increase clustering robustness. To choose the number of clusters, based on the trade-off between the number of clusters and the quality of the clusters, we used silhouette scores and knee point heuristics [70]. We followed the knee point heuristics to select the appropriate number of clusters using the algorithm from Satopaa et al., with a smoothing factor of 2 [73]. We did not consider results from more than 20 clusters, as it would be unrealistic for real-world readers to choose from this many reading themes.

D COR THEMES IN CSS

Table 2 shows the CSS values that make up COR themes.

	Compact	Open	Relaxed
characterSpacing (em)	0	0.02	0.04
wordSpacing (em)	0.1	0.2	0.3
lineHeight	1.4	2.2	4.5
fontName	Georgia	Merriweather	Poppins
fontSize (px)	15.8	15.8	14.1

Table 2: The three final themes' CSS values.

Iteration	Dyslexic	Spacing Setting	df	t	p	d
R0	Yes	Character	90	4.03	0.00	0.60
R0	Yes	Line	90	6.15	0.00	0.89
R0	Yes	Word	90	5.47	0.00	0.72
R0	No	Character	99	3.94	0.00	0.56
R0	No	Line	99	4.59	0.00	0.62
R0	No	Word	99	3.64	0.00	0.44
R1	Yes	Word	47	2.18	0.04	0.33
R2	No	Character	48	4.14	0.00	0.37
R3	Yes	Word	48	3.42	0.00	0.54
R3	No	Word	49	2.58	0.01	0.41

Table 3: Refinements from the reading themes made by participants with and without dyslexia, based on paired t-tests. Participants made significant refinements in all three spacing settings from the randomly initialized reading themes shown in the initial refinement iteration. The iterations that followed saw fewer adjustments in comparison. Degrees of freedom, t-statistics, p-value, and Cohen's d values are shown.

E CHANGES WITHIN ITERATIONS

Table 3 shows the number of statistical significant adjustments made throughout THERIF iterations.

F ADJUSTMENT TIME BY ITERATION

We applied Welch's t-test to compare adjustment times between R0 and R1 due to their unequal variances and sample sizes. We applied t-tests of equal variance for subsequent comparisons. Only time spent on refinement after the second round of "Review themes" step (§4.1) is considered adjustment time. See Figure 7 for comparison of adjustment time between iterations.

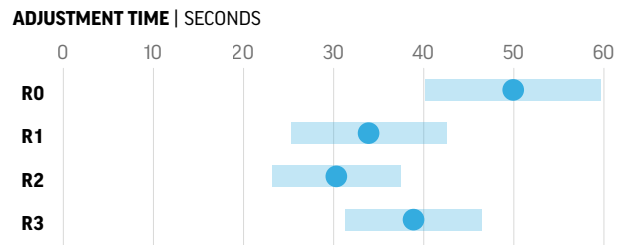


Figure 7: Average time participants spent adjusting text settings in each of the THERIF iterations. Compared to starting from a randomly initialized theme in R0, starting from a theme (R1-R3) reduced the average time participants spent on refining the text settings. Note that R3 had 50% fewer themes than previous iterations, but the refinement time did not change significantly.